Performance and cost

How can the same answer require less reading?

Paired activity · 20 minutes · Names: _____

Work with a partner. Take turns explaining the evidence before writing. All cases use the sales below; no computer is needed.

Date	Store	Units
2026-10-12	S03	5
2026-10-12	S04	1
2026-10-13	S03	3
2026-10-13	S04	9
2026-10-14	S03	6
2026-10-14	S04	2

Each row also has product STR. The same typed rows are available as sales.csv, sales.parquet, or date folders. Each date folder contains S03.parquet and S04.parquet, one row per file. The reader uses folder names, not file names, to skip files.

1. Two conditions, two kinds of evidence (7 minutes)

a. Find SUM(units) where date >= 2026-10-13 AND store = S03. Using only the date in each folder name, which files must still be opened? Which rows then match, and what is the total? If the same query reads sales.parquet instead, which columns must it read?

b. In a separate single Parquet file, chunk A spans October 12–13 and chunk B spans October 14. For the same query, can either chunk be skipped using only its minimum and maximum date? Can those date summaries establish store = S03? Explain.

A partition folder records a key's value. A row group is a chunk of Parquet rows. A zonemap summarizes a chunk's minimum and maximum values. A range containing a possible match cannot rule the chunk out. Projection pushdown reads only needed columns.

Explain the work

2. A store-only total (5 minutes)

The request is now SUM(units) where store = S03, with no date condition. The only folder key is date. Start with the six sales rows on page 1.

This conceptual scan/filter/sum model separates reading from filtering. Its counts describe rows produced by each step, not bytes read.

Step	EXPLAIN: before running	EXPLAIN ANALYZE: after running
Scan: read all sales rows	~6 rows	6 rows
Filter: keep store S03	~2 rows	Fill in below
Sum: combine units	~1 row	Fill in below

a. Complete the executed row counts for filter and sum. Write the values that enter the sum and its result. Which estimate differs from the actual count?

b. Your partner says, “The date folders skip files because only S03 matches, and the one-row total proves very little data was read.” Assess both claims using the folder names and the plan counts.

Reading the evidence

EXPLAIN shows a plan without running the query; ~ marks an estimate. EXPLAIN ANALYZE runs the query and reports actual row counts and time. Scan means read; filter means keep matching rows; aggregate means combine values.

For timing comparisons: query time excludes setup and downloads. A faster repeat can reflect cached data. Do not add operator times across parallel work. Matching rows, files and bytes measure different things.

Pay for useful work

3. More workers or better division? (3 minutes)

A larger sales snapshot assigns 120 equal-cost rows to simultaneous workers A, B, C and D as 60, 20, 20 and 20. One row takes one time unit. Option X adds idle worker E but leaves assignments unchanged. Option Y keeps four workers and redistributes rows to 30 each. Give each option's finish time and explain the difference. If sales and store records are on different workers, why might joining them still add work?

4. A correction splits the reuse window (5 minutes)

CSV takes 3 seconds per query. Preparing a layout costs 10 seconds, then queries take 1 second each. Run three queries, receive a corrected snapshot that requires rebuilding the prepared layout, then run three more queries. Compare total elapsed work for these six queries. These are example costs, not measured timings.

a. Show both totals, including every preparation, and choose a route on elapsed time. Why could treating all six queries as one unchanged batch give the wrong choice? Name one memory or storage cost beyond elapsed time.

b. Separately, a saved total covers October 13 only. The correction changes S04's October 12 units from 1 to 4; every October 13 row stays unchanged. Can the October 13 answer be reused? State the freshness rule that justifies your choice.